Assignment 2 CA-Software

E - Commerce Platform: Microservices & Event-Driven Architecture.

Microservices Architecture :

The e-commerce platform is decomposed into domain specific, independent microservices:

- Product Service :- Manages product catalog & inventory.
- Order Service :- Handles order creation, updates & cancellations.
- Payment Service : Process secure customer payments.
- Shipping Service :- Oversees delivery & logistics.
- Notification service :- Sends updates on orders, promotions etc.
- Customer service :- Manages profiles, authentication, & preferences.

Each service uses its own database & communicates via APIs or message queues, ensuring loose coupling & ~~scal~~ scalability.

Event-Driven Architecture (EDA)

Services communicate ~~asynchronously~~ asynchronously using events:-

1> OrderPlaced -> Emitted by Order Service when an order is placed.
2> PaymentProcessed -> Emitted by payment Service after processing payment.
3> ShippingStarted -> Triggered by Shipping service after payment is confirmed.
4> Notifications -> sent by Notification Service based on relevant events

SOLID Principles Applied :

- SRP : Each microservice has one responsibility (e.g., Product Service manages products only).
- OCP : New features (e.g., promotions) can be added without changing existing code.
- LSP : Services can be upgraded without breaking the system.
- ISP : Services expose only specific, necessary APIs.
- DIP : High level services rely on abstractions like events, not specific implementations

DRY & KISS Principles:

- DRY:-
  - Use shared frameworks for event processing.
  - Share common libraries (e.g., validation, auth) across services.

- KISS:-
  - Keep services focused & simple.
  - Design clear, self-contained events.
  - Minimized inter-service dependencies.